Languages:

Python

Databases:

CSV/Parquet Flat File Systems

Tools:

PyCharm / VS Code, **Jupyter Notebook**, **GitHub**, **Jira**

Technologies:

Pandas, **NumPy**, **JSON Processing**, **Python File Handling (os, pathlib, shutil)**,
Regular Expressions (Regex), **Modular Programming**, **User-Defined Functions (UDFs)**,
Error Handling (try–except), **Data Validation Frameworks**, **ETL/ELT Pipeline Development**, **Data Preprocessing**, **Data Transformation**, **Metadata-Driven Processing (Control Files)**,
Parquet File Processing, **CSV Ingestion**, **Audit Logging**,
Data Quality Checks, **Row-Count & Schema Reconciliation**,
Pipeline Orchestration, **Aggregations & KPIs using Pandas**, **GroupBy Aggregations**,
Python Packaging & Modular Design, **File Archival & Retention (ZIP compression)**,
Source-to-Target Mapping (STTM), **Data Lineage & Traceability**

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Roles & Responsibilities:

• Developed end-to-end **Python-based ETL workflows** for **RAW, PREPROCESSED, CURATED, and SEMANTIC layers** using **modules, packages, UDFs, control structures, file handling**, and **Pandas-based data pipelines** to automate ingestion, validation, preprocessing, transformation, and archival activities.

• Implemented **control-file–driven ingestion** using **JSON control file**, **Python dictionaries**, and custom **User-Defined Functions (UDFs)** to perform **schema validation**, **file existence checks**, **expected column validation**, **mandatory field checks**, **file-date extraction**, **datatype verification**, and **row-count reconciliation**—ensuring complete integrity before onboarding datasets.

• Utilized **Pandas (DataFrames)**, **Python built-in functions**, and **vectorized operations** for high-performance ingestion of **CSV files** including **delimiter handling**, **UTF encoding**, **dtype assignment**, **header validation**, and conversion to optimized **Parquet format**.

• Designed preprocessing logic using **Python control structures (if-else, loops)**, **list comprehensions**, **UDFs**, and **Pandas transformations** to perform **deduplication**, **null**

standardization, **trimming**, **lower/upper case conversion**, **derived field generation**, **timestamp stamping**, and **feature enrichment** before loading data into curated layers.

• Built curated transformation pipelines using **Pandas joins**, **merge operations**, **dictionary-based mappings**, **column alignment logic** to integrate **Claims, Customers, Policies, and Payments** dataframe.

• Engineered **semantic-level datasets** using **groupby aggregations**, **KPI derivation**, **functional programming (lambda, map)**, **statistical summaries**, **pivot/unpivot logic**, and exported them using **Pandas I/O** and **file system utilities** into **Parquet/CSV** file formats.

• Created a robust **audit logging framework** using **Python file handling**, **CSV writer**, **datetime utilities**, and custom **UDFs** to consistently capture **ingestion metadata**, **timestamps**, **row counts**, **error messages**, and **execution status** for full audit traceability.

• Implemented **exception handling (try–except)**, **custom error propagation**, and **safe-exit mechanisms** across ingestion, validation, preprocessing, and transformation layers to ensure pipeline stability and predictable error reporting.

• Designed **archival and retention logic** using **os module**, **shutil**, **pathlib**, and **ZIP compression utilities**, automating the movement of processed files and JSON control files into **timestamped archive directories** for compliance and lineage preservation.

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• Wrote reusable **User-Defined Functions (UDFs)** for **filename date extraction**, **timestamp generation**, **column cleaning**, **schema verification**, **metadata parsing**, and **standardized transformation rules**—ensuring modularity and maintainability across pipeline components.

• Collaborated with **Data Engineers**, **BAs**, and **QA teams** to convert business logic into **Python transformation rules**, **Pandas operations**, **JSON-driven mappings**, and domain-specific curated models.